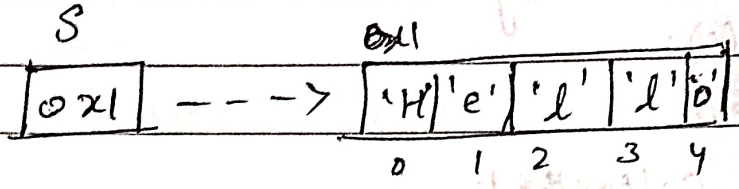
Strings

* Group of characters enclosed within double quotes. is called as String.

* Each and every string that we create will stored in the form of a character Array.

String s = "Hello";



* charAt(index);

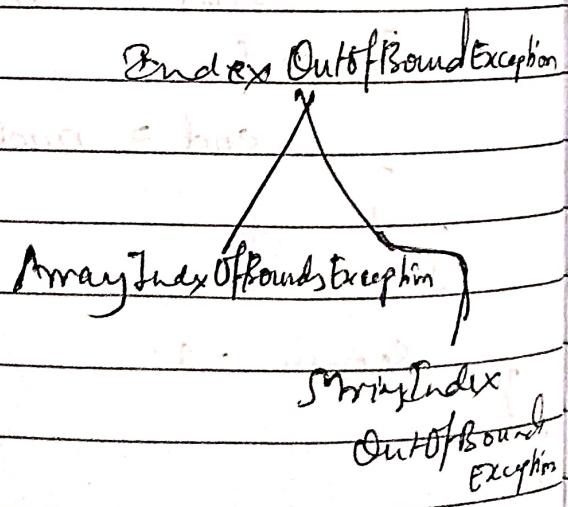
Eg: • Sopln(s.charAt(2)); // l

• Sopln(s.charAt(5)); // StringIndexOutOfBounds Exception

* length()

Eg: Sopln(s.length()); // 5

* next()
nextLine()



① Program to read String data from the user using next & nextLine(). & print the concatenation of both Strings.

```
Scanner sc = new Scanner(System.in);
Sopln("Enter a word: ");
String a = sc.next();
sc.nextLine();
Sopln("Enter a sentence: ");
String b = sc.nextLine();

S.O.P(a + " " + b);
```

② Program to print the characters from string

```
String str = "hello";

for(int i=0; i<str.length(); i++){
        S.o.p(str.charAt(i) + " ");
}
```

③ Program to print only vowels in a given string

```
String str = "hello";

for(int i=0; i<str.length(); i++){
    char ch = s.charAt(i);
    if(ch=='A' || ch=='E' || ch=='I' || ch=='O'
    || ch=='U' || ch=='a' || ch=='e' || ch=='i' ||
        ch=='o' || ch=='u"){
        S.O.P(ch);
}
```

(21) Program to print the count of vowels & count of constants in a given string.

```
int v=0;    int c=0;
String str = "Hello";

for (int i=0; i< str.length(); i++) {
    Char ch = str.charAt(i);

    if ( ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' ||
         ch == 'U' || ch == 'a' || ch == 'e' ||
         ch == 'i' || ch == 'o' || ch == 'u' ){
            v++;
    }
    else {
        c++;
    }
}
S.o.p (v);
S.o.p (c);
```

(5) Print the count of characters in a given string excluding spaces.

```
String str = "Hello world";
int count = 0;

for (int i=0; i< str.length(); i++) {
        if (str.charAt(i) != ' ') {
            count++;
        }
}
Sopln(count);
```

(6) Program to print count of characters in given string without using length variable & length() method;

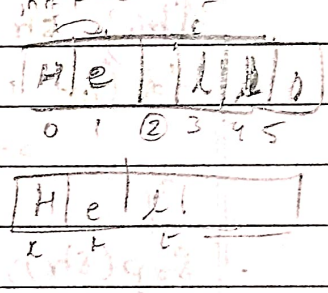
```
int count = 0;
String str = "Hello world";
char[] ch = str.toCharArray();
for (char c : ch) {
        count++;
}
S.O.P (count);
```

for-each loop cannot be used with ~~character~~ & string directly.

To use for foreach on strings convert to charArray()

Note: Using for-each we can only access values, we cannot modify,
If we try to modify we get ConcurrentException

(7) Program to remove spaces from given string




```
String s1 = "Hello Good Morning";
String s2 = "";
for (int i=0; i< s1.length(); i++) {
    if (s1.charAt(i) != ' ')
        {
            s2 = s2 + s1.charAt(i);
        }
}
S.O.P (s2);
```

(8) Program to sort the characters in given string in ascending order.

H	e	l	l	o

```
String s1 = "Hello";
char[] ch = s1.toCharArray();

for(int i=0; i<ch.length; i++){

    for(int j=i+1; j<ch.length ; j++){

        if(ch[i] > ch[j]){
            char temp = ch[i];
            ch[i] = ch[j];
            ch[j] = temp;
        }
    }
}
```

~~System.out.println(ch);~~ Sop(new String(ch));

(or)

```
String str="";
for(int i=0; i<=ch.length-1; i++){
    str = str+ ch[i];
}
Sop(str);
```